Maximal recoupling, and simultaneously maximal coupling under independence constraints

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Written up from the research notes in October 2026 with AI assistance; see Section 10.

October 2026

Abstract

Given a joint distribution of two random variables X and Y on a finite set, the classic maximal coupling algorithm samples a new joint distribution that keeps both marginals and maximizes $\mathbb{P}[X = Y]$. It discards the structure of the original joint, and in particular it destroys subcouplings that were already maximal. We present *maximal recoupling* (MRC), a small generalization of the classic rejection-sampling algorithm. MRC turns an arbitrary coupling into a maximal one, and every coarsening $(f(X), f(Y))$ that was maximally coupled before comes out with exactly the same joint law. More quantitatively, MRC never decreases $\mathbb{P}[f(X) = f(Y)]$, and it moves the law of $(f(X), f(Y))$ by at most twice the improvement in total variation. MRC needs only likelihoods of the two marginals and a conditional sampler for the original coupling, and it makes at most one expected call to that sampler per sample. We give a closed form for its output and a decomposition of its effect into shortcut random-walk trajectories.

As the application, we study *simultaneously maximal coupling*: coupling $X = (X_i)_{i \in V}$ and $Y = (Y_i)_{i \in V}$ so that X_S and Y_S are maximally coupled for every S in a family $\mathcal{F}$. Assume every coordinate takes at least two values. Without further assumptions this is possible for all distributions exactly when $\mathcal{F}$ is a chain. This nested case is known and underlies block verification in speculative decoding. If the coordinates of X are independent and so are those of Y, it is possible exactly when $\mathcal{F}$ is laminar. If X and Y are Bayesian networks on a common DAG, it is possible exactly when $\mathcal{F}$ is laminar and its disjoint members have disjoint ancestor sets. The constructive direction is an efficient sampling algorithm on *chain-product trees*, built from MRC. In the autoregressive special case it reproduces the known all-prefix maximal coupling.

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1 Introduction

A *coupling* of two distributions μ and ν is a joint distribution of a pair (X, Y) with $X \sim \mu$ and $Y \sim \nu$. The coupling inequality says $\mathbb{P}[X \neq Y] \geq d_{\mathrm{TV}}(\mu, \nu)$, and a coupling that achieves equality is called *maximal*. Maximal couplings go back to Doeblin [9]. There is a classic rejection-sampling algorithm for drawing from one using only samplers and likelihoods for μ and ν [6].

The same algorithm, in disguise, is the token-level verification step of speculative decoding for large language models [8, 2, 14]. This research started from a related use of coupling: comparing two generative models (for example, a model and its quantization) by generating from them jointly so that their responses agree as often as possible. Unforced differences then stand out against a background of shared randomness. Two observations drive the rest:

- Coupling each next-token distribution maximally ("per-token optimal coupling") does not maximally couple whole responses, or even prefixes of responses (Section 7). It *is* possible to couple every prefix maximally at once. This was found independently by the author and, earlier and in other contexts, by others [16, 3, 15].
- Generative models whose sampling involves conditionally independent choices, such as discrete-diffusion language models that unmask several positions independently in one step [11], raise a more general question: which families of subsets of the coordinates can be maximally coupled *simultaneously*, and how do we sample from such a coupling?

Contributions.

1. **Maximal recoupling (Sections 3 and 4).** MRC (Algorithm 2) has the same two-line shape as the classic algorithm. The difference is that the "try again" draws from Y are replaced by a random walk along the conditional law $P(Y \mid X)$ of the input coupling.

Theorem 3.2 proves that the output is a maximal coupling, preserves every maximal coarsening exactly, never decreases the agreement probability of any coarsening, and costs at most one expected call to $P(Y \mid X)$ per sample. The cost bound propagates through nested use. If inputs arrive with expected frequencies at most p_X, then the old sampler is called with expected frequencies at most p_X, and outputs occur with expected frequencies at most p_Y. This is what lets MRC be nested. Proposition 3.4 gives the closed form. Every argument needed about Markov chains is elementary and confined to one structural lemma (Lemmas 4.1 and 4.2).

2. **Simultaneously maximal coupling on chain-product trees (Section 5).** A chain-product tree (CPT) describes a distribution built from atomic variables by chain-rule extensions and independent products. The SMC-CPT algorithm applies MRC at every node. Its output is maximal on the scope of every node (Theorem 5.3), and each Y-side oracle is called at most once in expectation.

3. **Characterizations (Section 6).** Under no structural assumption, a family can be simultaneously maximally coupled for all distributions iff it is a chain (Theorem 6.2). Under independence, iff it is laminar (Theorem 6.3). For Bayesian networks on a DAG G, iff it is laminar and its disjoint members have disjoint ancestor sets (Theorem 6.5). If every member is also an ancestral set, the CPT oracles are just the network's conditional distributions (Corollary 6.6).

4. **The autoregressive case (Section 7).** On a pure chain, SMC-CPT outputs exactly the "all-prefix-optimal coupling" (APOC) defined by an explicit greedy formula (Theorem 7.2). This settles a conjecture from the notes. We also record that per-token maximal coupling dominates every seed-reuse scheme prefix-by-prefix (Proposition 7.3), along with small examples.

What is new? The nested (chain) case is not new. It is the setting of maximal agreement couplings [16, 3] and of block verification in speculative decoding [15]. Griffeath's and Goldstein's classical maximal couplings of processes [5, 4] handle the reverse nesting, of tail σ-fields. The author searched in 2024 and did not find MRC or the laminar and Bayesian-network results in the literature. A later search, done when this write-up was planned, also found nothing. Literature searches are never exhaustive, so treat novelty claims here as provisional.

Reading guide. Section 2 fixes notation. Section 3 states MRC and its properties, and Section 4 proves them. Section 5 gives the simultaneous construction and Section 6 the characterizations. Section 7 treats autoregressive models. Sections 8 and 9 cover related work and open problems. Appendix A maps the terminology of the original notes to the terminology used here, and Appendix B gives implementation-level pseudocode.

2 Preliminaries

Throughout, E is a finite set. A distribution P on $E \times E$ is identified with the matrix $(P(x,y))_{x,y \in E}$. For $S, T \subseteq E$ we write $P(S,T) = \sum_{x \in S,\, y \in T} P(x,y)$. Its marginals are $p_X(x) = P(\{x\}, E)$ and $p_Y(y) = P(E, \{y\})$, and $(X,Y) \sim P$ denotes a random pair with law P. A *coupling* of distributions

μ, ν on E is a P with $p_X = \mu$ and $p_Y = \nu$. We write $a^+ = \max\{a, 0\}$ and

$$d_{\mathrm{TV}}(\mu, \nu) = \sum_x (\mu(x) - \nu(x))^+ = \max_{A \subseteq E} (\mu(A) - \nu(A)).$$

We use the same formula for the distance between two distributions on $E \times E$.

Definition 2.1 (maximal coupling)**.** A coupling P is *maximal* if $P(x, x) = \min\{p_X(x), p_Y(x)\}$ for every $x \in E$.

Since $P(x, x) \le \min\{p_X(x), p_Y(x)\}$ always holds, and $\sum_x \min\{p_X(x), p_Y(x)\}$ equals $1 - d_{\mathrm{TV}}(p_X, p_Y)$, a coupling is maximal iff $\mathbb{P}[X = Y] = 1 - d_{\mathrm{TV}}(p_X, p_Y)$, the largest possible value.

For a coupling P define the *sources*, *sinks* and *surpluses*

$$E_+ = \{x : p_X(x) > p_Y(x)\}, \qquad E_- = \{y : p_Y(y) > p_X(y)\},$$
$$\alpha(x) = (p_X(x) - p_Y(x))^+, \qquad \beta(y) = (p_Y(y) - p_X(y))^+.$$

Picture P as a flow on the directed graph with an arc $x \to y$ of weight $P(x, y)$ for each pair with $P(x, y) > 0$, self-loops included. A node's out-weight is p_X and its in-weight is p_Y. Sources have excess out-weight α and sinks excess in-weight β, and $\sum \alpha = \sum \beta = d_{\mathrm{TV}}(p_X, p_Y)$.

Lemma 2.2 (local characterization)**.** *A coupling P is maximal iff no $x \in E$ has both $P(x, E \setminus \{x\}) > 0$ and $P(E \setminus \{x\}, x) > 0$. If P is maximal and $P(x, y) > 0$ with $x \neq y$, then $x \in E_+$ and $y \in E_-$.*

Proof. $P(x, E \setminus \{x\}) = p_X(x) - P(x, x)$ and $P(E \setminus \{x\}, x) = p_Y(x) - P(x, x)$. Both are positive iff $P(x, x) < \min\{p_X(x), p_Y(x)\}$. If P is maximal and $P(x, y) > 0$ with $y \neq x$, then $P(x, E \setminus \{x\}) > 0$, so $P(E \setminus \{x\}, x) = 0$. Hence $P(x, x) = p_Y(x)$ and $p_X(x) = p_Y(x) + P(x, E \setminus \{x\}) > p_Y(x)$. The statement for y is symmetric. □

Coarsenings. For a map $f \colon E \to E'$, the *coarsening* P^f is the joint law of $(f(X), f(Y))$: $P^f(a, b) = P(f^{-1}(a), f^{-1}(b))$. It is a coupling of the pushforwards of p_X and p_Y. We say P is *maximal on* f if P^f is a maximal coupling. Equivalently, take the partition $\mathcal{F}_f$ of E into the nonempty fibres of f. Then P^f is the block matrix $(P(S, T))_{S,T \in \mathcal{F}_f}$ and $\mathbb{P}[f(X) = f(Y)] = \sum_{S \in \mathcal{F}_f} P(S, S)$. The main example is $E = \prod_{i \in V} E_i$ with f the projection $x \mapsto x_S$ onto a subset of coordinates.

Oracle model. Algorithms access distributions only through oracles. A *likelihood oracle* for μ returns $\mu(x)$ given x. A *sampler* for μ returns an independent draw from μ on each call. A *conditional sampler* for a coupling P takes v with $p_X(v) > 0$ and returns an independent draw from $P(\cdot \mid v) = P(v, \cdot)/p_X(v)$, the conditional law of Y given $X = v$. Each oracle call uses fresh randomness, independent of everything else. Coin flips "with probability q" are free.

Let $D = d_{\mathrm{TV}}(p_X, p_Y)$. If $D = 0$, CMC-Cond returns its input, and the coupling is the diagonal one. If $D > 0$, drawing $x \sim p_X$ and then $y \leftarrow \text{CMC-Cond}(x)$ (Algorithm 1) gives the maximal coupling $\min\{p_X(x), p_Y(x)\}\mathbf{1}[x = y] + \alpha(x)\beta(y)/D$, and the returned residual y is independent of x [6, 10]. Once entered, the loop makes $1/D$ draws in expectation. It is entered with probability D, so there is exactly one expected draw per sample.

Algorithm 1 Classic maximal coupling, conditional form (CMC-Cond)

Input: x with $p_X(x) > 0$
Access: likelihoods p_X, p_Y; sampler for p_Y

with probability $\min\{1, p_Y(x)/p_X(x)\}$: **return** x
loop
  draw $y \sim p_Y$
  with probability $\min\{1, p_X(y)/p_Y(y)\}$: **continue**
  return y ▷ reached with probability $\beta(y)/p_Y(y)$

Algorithm 2 Maximal recoupling, conditional form (MRC-Cond$_P$)

Input: x with $p_X(x) > 0$
Access: likelihoods p_X, p_Y; conditional sampler for $P(\cdot \mid \cdot)$

with probability $\min\{1, p_Y(x)/p_X(x)\}$: **return** x
$v \leftarrow x$
loop
  draw $w \sim P(\cdot \mid v)$
  with probability $\min\{1, p_X(w)/p_Y(w)\}$: $v \leftarrow w$; **continue**
  return w ▷ reached with probability $\beta(w)/p_Y(w)$

3 Maximal recoupling

Maximal recoupling of a coupling P draws $x \sim p_X$, then $y \leftarrow \text{MRC-Cond}_P(x)$ (Algorithm 2), and outputs (x, y). We write $\hat{P} = \text{MRC}(P)$ for the law of the output and call it the *recoupling* of P. The only change from Algorithm 1 is in the loop: a fresh draw from p_Y becomes a draw from $P(\cdot \mid v)$, where v is the most recent rejected value. A rejected Y-value is thus reused as an X-value. The loop is a random walk that follows the flow P from the source x until it is absorbed at a sink.

Remark 3.1 (MRC generalizes CMC)**.** If $P = p_X \otimes p_Y$ is the independent coupling, then $P(\cdot \mid v) = p_Y$ for every v and Algorithm 2 is literally Algorithm 1.

Theorem 3.2 (properties of maximal recoupling)**.** *Let P be a coupling on E and $\hat{P} = \text{MRC}(P)$.*

(a) ***Termination.*** *For every x with $p_X(x) > 0$, MRC-Cond$_P(x)$ terminates almost surely.*

(b) ***Validity.*** *$\hat{P}$ is a coupling of p_X and p_Y.*

(c) ***Maximality.*** *$\hat{P}$ is maximal, and $\hat{P}(x, y) > 0$ with $x \neq y$ implies $x \in E_+$, $y \in E_-$.*

(d) ***Efficiency.*** *Suppose MRC-Cond$_P$ is invoked a random, possibly adaptively chosen, number of times, and let $c(x)$ be the expected number of invocations with input x. If $c(x) \leq \lambda\, p_X(x)$ for all x, for some $\lambda \geq 0$, then for every v the expected number of calls to the conditional sampler with input v is at most $\lambda\, p_X(v)$, and for every y the expected number of invocations that return y is at most $\lambda\, p_Y(y)$. In particular, one run of MRC makes at most one expected call to the conditional sampler.*

(e) ***Diagonal blocks do not shrink.*** *$\hat{P}(S, S) \geq P(S, S)$ for every $S \subseteq E$.*

(f) ***Off-diagonal gains are paid for.*** *For every partition $\mathcal{F}$ of E,*

$$\sum_{\substack{S,T\in\mathcal{F}\\ S\neq T}} \big(P(S,T)-\hat{P}(S,T)\big) \;\geq\; \sum_{\substack{S,T\in\mathcal{F}\\ S\neq T}} \Big(\hat{P}(S,T)-P(S,T)\Big)^{+}.$$

Corollary 3.3 (effect on coarsenings)**.** *Let $f\colon E\to E'$ and write $\gamma_f=\hat{P}[f(X)=f(Y)]-P[f(X)=f(Y)]$. Then:*

(i) $\gamma_f\geq 0$: recoupling never lowers the agreement probability of a coarsening.

(ii) $d_{\mathrm{TV}}(P^f,\hat{P}^f)\leq 2\gamma_f$.

(iii) If P is maximal on f, then $\hat{P}^f=P^f$. In particular, $\hat{P}$ is maximal on f.

Part (iii) applies to all maximal coarsenings of P at once. That is what makes MRC the right building block for simultaneous coupling (Section 5). Part (ii) is a quantitative version: a coarsening that was nearly maximal moves only a little. The constant 2 is sharp (Example 3.7).

Proposition 3.4 (closed form)**.** *Let $p_{\max}(v)=\max\{p_X(v),p_Y(v)\}$, and let A be the $E\times E$ matrix*

$$A_{vw}=\frac{P(v,w)}{\sqrt{p_{\max}(v)\,p_{\max}(w)}} \quad \text{if } p_{\max}(v)p_{\max}(w)>0, \qquad A_{vw}=0 \text{ otherwise.}$$

Then $\hat{P}(x,x)=\min\{p_X(x),p_Y(x)\}$. For $x\neq y$, $\hat{P}(x,y)=0$ unless $x\in E_+$ and $y\in E_-$, in which case

$$\hat{P}(x,y)=\frac{\alpha(x)\,\beta(y)}{\sqrt{p_X(x)\,p_Y(y)}}\,\big[(I-A)^{+}\big]_{xy}=\frac{\alpha(x)\,\beta(y)}{\sqrt{p_X(x)\,p_Y(y)}}\sum_{n\geq 1}(A^n)_{xy},$$

where $(I-A)^{+}$ is the Moore–Penrose pseudoinverse and the series converges. If every node v with $p_{\max}(v)>0$ can reach a sink in the flow graph of P (by a path of length ≥ 0), then $I-A$ is invertible and $(I-A)^{+}=(I-A)^{-1}$. In general $(I-A)^{+}$ may be replaced by $(I-A_R)^{-1}$, where A_R is the restriction of A to the set R of nodes reachable from a source.

When $P=p_X\otimes p_Y$, the sum is a geometric series with ratio $1-d_{\mathrm{TV}}(p_X,p_Y)$, and the formula reduces to the CMC formula of Section 2.

Corollary 3.5 (symmetry)**.** $\mathrm{MRC}(P^{\top})=\mathrm{MRC}(P)^{\top}$*, where $P^{\top}(x,y)=P(y,x)$. Running* MRC *with the roles of X and Y exchanged gives the transposed coupling.*

Proof. Transposing P exchanges p_X with p_Y, α with β and E_+ with E_-, and replaces A by $A^{\top}$. The formula of Proposition 3.4 is invariant under these changes together with $(x,y)\mapsto(y,x)$, since $(A^{\top})^n_{yx}=A^n_{xy}$. □

3.1 Examples

Example 3.6 (why recoupling: CMC destroys maximal subcouplings)**.** Let $E=\{0,1\}^2$. Draw X_1 uniform, X_2 uniform and independent of X_1, and set $Y=(X_1,0)$. Then (X_1,Y_1) is maximally coupled, since $X_1=Y_1$ always. Applying CMC to the marginals of X and Y keeps $X=Y$ when $X_2=0$. When $X_2=1$, it draws the residual $Y\sim\mathrm{Unif}\{(0,0),(1,0)\}$ independently of X, so $\mathbb{P}[X_1\neq Y_1]=1/4$. MRC instead walks from $x=(x_1,1)$ to $w\sim P(\cdot\mid x)=\delta_{(x_1,0)}$, which is a sink, and returns it. So $X_1=Y_1$ is preserved, as Corollary 3.3(iii) promises.

Example 3.7 (the constant 2 is sharp)**.** Let P be uniform on $\{(a,b),(c,a)\}$, so $X \sim \mathrm{Unif}\{a,c\}$ and $Y \sim \mathrm{Unif}\{a,b\}$. The unique maximal coupling is uniform on $\{(a,a),(c,b)\}$, and MRC produces it: from the source c it walks to a, continues, walks to b, and stops. With $f = \mathrm{id}$, $d_{\mathrm{TV}}(P,\hat{P}) = 1$ and $\gamma_f = 1/2$.

Example 3.8 (the lower bound $\gamma_f \le d_{\mathrm{TV}}$ is sharp)**.** In general $\gamma_f \le \sum_S \left(\hat{P}(S,S) - P(S,S)\right)^+ \le d_{\mathrm{TV}}(P^f, \hat{P}^f)$. For P uniform on $\{(a,b),(b,a)\}$, $\hat{P}$ is uniform on $\{(a,a),(b,b)\}$, and $d_{\mathrm{TV}}(P,\hat{P}) = \gamma_{\mathrm{id}} = 1$.

Example 3.9 (without recoupling, a maximal coupling can move by 1)**.** Let $X \sim \mathrm{Unif}\{a,b\}$ and $Y \sim \mathrm{Unif}\{c,d\}$. The couplings uniform on $\{(a,c),(b,d)\}$ and uniform on $\{(a,d),(b,c)\}$ are both maximal (here $\mathbb{P}[X=Y]=0$) and are at total variation distance 1 from each other. A procedure that only looks at the marginals can thus move a coupling that is already maximal by the largest possible amount. By Corollary 3.3(iii), MRC leaves any maximal coupling unchanged.

Example 3.10 (the properties do not determine $\hat{P}$)**.** Let P be uniform on $\{(a,c),(b,c),(c,d),(c,e)\}$. Then a, b are sources, d, e are sinks, and c is balanced. MRC gives $\hat{P}(c,c) = \frac{1}{2}$ and $\hat{P}(s,t) = \frac{1}{8}$ for $s \in \{a,b\}$, $t \in \{d,e\}$. The maximal couplings

$$\tfrac{1}{2}\delta_{(c,c)} + \tfrac{1}{4}\delta_{(a,d)} + \tfrac{1}{4}\delta_{(b,e)} \quad \text{and} \quad \tfrac{1}{2}\delta_{(c,c)} + \tfrac{1}{4}\delta_{(a,e)} + \tfrac{1}{4}\delta_{(b,d)}$$

satisfy Theorem 3.2(e),(f) for every S and every partition as well. Indeed, the first one is obtained by decomposing P into the paths $a \to c \to d$ and $b \to c \to e$, each of weight $\frac{1}{4}$, and shortcutting them. The second uses $a \to c \to e$ and $b \to c \to d$. In each case the weighted g- and $\hat{g}$-matrices of Section 4 sum to P and to the candidate, and the proofs of (e) and (f) go through verbatim. MRC is their average.

3.2 A short non-constructive existence proof

The existence of a maximal coupling that preserves all maximal coarsenings has a short proof by linear programming. It was in fact found before MRC. It gives no sampling algorithm. We give the finite-dimensional argument, although a compactness argument extends it to countable E. It shows the mechanism: the *exchange move*.

Proposition 3.11. *Let P be a coupling on E and let $f_1, \dots, f_m$ be maps on E such that P is maximal on each f_k. Then there is a maximal coupling P^* of p_X, p_Y with $(P^*)^{f_k} = P^{f_k}$ for all k.*

Proof. Maximize $\sum_x Q(x,x)$ over the polytope of distributions Q on $E \times E$ with marginals p_X, p_Y and $Q^{f_k} = P^{f_k}$ for all k. The polytope is nonempty since it contains P, and it is compact, so a maximizer P^* exists. Suppose P^* is not maximal. Write e_{vw} for the matrix unit. By Lemma 2.2 there are a, $b \neq a$ and $c \neq a$ with $\delta = \min\{P^*(a,b), P^*(c,a)\} > 0$. Let $\tilde{P} = P^* + \delta(e_{aa} + e_{cb} - e_{ab} - e_{ca})$. This has the same marginals, and its objective is larger by at least δ (by 2δ if $b = c$). Fix k and write $f = f_k$. Since $(P^*)^f = P^f$ is maximal and has mass on $(f(a), f(b))$ and $(f(c), f(a))$, Lemma 2.2 forces $f(a) = f(b)$ or $f(c) = f(a)$. In either case the move changes $(P^*)^f$ by $\delta(e_{f(a)f(a)} + e_{f(c)f(b)} - e_{f(a)f(b)} - e_{f(c)f(a)}) = 0$. So $\tilde{P}$ is feasible, contradicting optimality. □

4 Proofs for maximal recoupling

Fix a coupling P on E. Let $E_X = \{v : p_X(v) > 0\}$. For $v \in E_X$ let $K(v, w) = P(v, w)/p_X(v)$ be the conditional law used on line 4 of Algorithm 2. For w with $p_Y(w) > 0$ let

$$\rho(w) = \min\{1, p_X(w)/p_Y(w)\} = p_X(w)/p_{\max}(w)$$

be the continuation probability on line 5, and set $\rho(w) = 1$ if $p_Y(w) = 0$. Only w with $p_Y(w) > 0$ are ever drawn, because $K(v, w) = 0$ otherwise, so this convention is never used. For w with $p_Y(w) > 0$, $\rho(w) > 0$ iff $w \in E_X$, and $\rho(w) < 1$ iff $w \in E_-$. All sums over inputs or departure points below run over E_X. Quantities indexed by $v \notin E_X$ are 0.

4.1 The walk

Each iteration of the loop starts at the current value v. We say it *departs* from v, makes one call to the conditional sampler with input v, and *traverses* the arc (v, w) that it draws. It then either continues at w (probability $\rho(w)$) or exits, returning w. The loop is therefore a Markov chain on E_X with substochastic transition matrix $Q(v, w) = K(v, w)\rho(w)$ and exit probabilities $H(v, w) = K(v, w)(1 - \rho(w))$. From the input x, the loop is entered with probability $1 - \min\{1, p_Y(x)/p_X(x)\} = \alpha(x)/p_X(x)$. In one run of MRC, with $x \sim p_X$, the loop is therefore entered at x with probability $\alpha(x)$.

For $x \in E_X$, let $N_x(v) \in [0, \infty]$ be the expected number of departures from v during MRC-Cond$_P(x)$, and let $\mu(v) = \sum_{x \in E_X} p_X(x) N_x(v)$ be the expected number in one run of MRC. If $x \notin E_+$ the loop is never entered and $N_x = 0$. For $x \in E_+$, counting visits step by step gives

$$N_x = \frac{\alpha(x)}{p_X(x)} \sum_{n \geq 0} e_x^\top Q^n, \qquad \text{hence} \qquad \mu = \sum_{n \geq 0} \alpha^\top Q^n \quad \text{(row vectors on } E_X\text{)},$$

with α supported on E_+. The series are finite by Lemma 4.2. Each departure from v traverses (v, w) with probability $K(v, w)$ and exits at w with probability $H(v, w)$, independently of the past. So the expected number of traversals of (v, w) in one run is $\mu(v)K(v, w)$, and the expected number of exits at w is $\sum_v \mu(v)H(v, w)$.

4.2 Structure of the flow

Let G be the directed graph on E with an arc $v \to w$ iff $P(v, w) > 0$. Let R be the set of nodes reachable in G from E_+ (by paths of length ≥ 0), and let $U = E \setminus R$.

Lemma 4.1 (structure)**.**

(i) $P(R, U) = 0$ *and* $P(U, R) = 0$*.*

(ii) $p_X(u) = p_Y(u)$ *for every* $u \in U$*. In particular* $E_+ \cup E_- \subseteq R$*.*

(iii) Every $v \in R \cap E_X$ *has a path of positive length in* G *to a node of* E_-*.*

Proof. (i,ii) R is closed under following arcs, so $P(R, U) = 0$. Then

$$p_X(U) - p_Y(U) = P(U, R) + P(U, U) - P(R, U) - P(U, U) = P(U, R) \geq 0.$$

But $U \cap E_+ = \emptyset$, so $p_X(u) \leq p_Y(u)$ for every $u \in U$. Hence $P(U, R) = 0$ and $p_X = p_Y$ on U.

(iii) Since $p_X(v) > 0$ there is w with $P(v,w) > 0$, and $w \in R$. Let J be the set of nodes reachable from w, including w. Suppose $J \cap E_- = \emptyset$. J is closed, so $P(J, E \setminus J) = 0$ and $p_Y(J) - p_X(J) = P(E \setminus J, J) \geq 0$. With no sinks in J, $p_Y \leq p_X$ on J, so $P(E \setminus J, J) = 0$ and $p_X = p_Y$ on J. In particular $J \cap E_+ = \emptyset$. But $w \in R$, so a path leads from some $s \in E_+$ to w. Since $s \notin J$, this path enters J through an arc with positive weight from $E \setminus J$ into J, which is a contradiction. Thus some $u \in J \cap E_-$ is reachable from v by a path $v \to w \to \cdots \to u$ of positive length. □

Lemma 4.2 (the walk)**.** *Let $R_X = R \cap E_X$ and let Q_R be the restriction of Q to R_X.*

(i) Started in R_X, the walk stays in R_X until it exits.

(ii) The spectral radius of Q_R is less than 1. Hence the walk exits almost surely from every start in R_X, and the number of departures has finite expectation.

(iii) $\mu(v) = p_X(v)$ for $v \in R$, and $\mu(v) = 0$ for $v \in U$.

(iv) The expected number of traversals of (v,w) in one run of MRC *is $T(v,w) = P(v,w)\mathbf{1}[v \in R]$.*

Proof. (i) If $v \in R_X$ and $Q(v,w) > 0$, then $P(v,w) > 0$, so $w \in R$, and $\rho(w) > 0$, so $w \in E_X$.

(ii) Let $v \in R_X$. By Lemma 4.1(iii) there is a path $v = u_0 \to u_1 \to \cdots \to u_m$ with $m \geq 1$ and $u_m \in E_-$. Truncating at the first sink among $u_1, \ldots, u_m$, we may assume $u_1, \ldots, u_{m-1} \notin E_-$, and also that $m \leq |E|$. For $1 \leq i < m$ we have $p_Y(u_i) > 0$ (it has an in-arc) and $u_i \notin E_-$, so $p_X(u_i) \geq p_Y(u_i) > 0$ and $\rho(u_i) = 1$. The walk follows this path and exits at u_m with probability $\prod_{i=1}^{m} K(u_{i-1}, u_i)\,(1 - \rho(u_m)) > 0$. Since R_X is finite, there is $\varepsilon > 0$ such that from every state of R_X the walk exits within $|E|$ steps with probability at least ε. So every row sum of $Q_R^{|E|}$ is at most $1 - \varepsilon$, and $\|Q_R^{|E|}\|_\infty < 1$.

(iii) The loop is entered only at sources, and $E_+ \subseteq R_X$. By (i), μ is supported on R_X, and by (ii), $\mu|_{R_X} = \alpha|_{R_X}^\top (I - Q_R)^{-1}$ is the unique solution of $m = \alpha|_{R_X} + mQ_R$. We check that $m = p_X|_{R_X}$ is also a solution. For $w \in R_X$, nodes $v \in R \setminus E_X$ have $P(v, \cdot) = 0$, and $P(U, w) = 0$ by Lemma 4.1(i). Hence

$$\sum_{v \in R_X} p_X(v) Q(v,w) = \rho(w) \sum_{v \in R_X} P(v,w) = \rho(w)\, P(E,w) = \rho(w) p_Y(w) = \min\{p_X(w), p_Y(w)\},$$

and $\alpha(w) + \min\{p_X(w), p_Y(w)\} = p_X(w)$. So $\mu = p_X$ on R_X. On $R \setminus E_X$ both sides are 0, and on U we have $\mu = 0$.

(iv) The expected number of traversals is $\mu(v)K(v,w) = P(v,w)\mathbf{1}[v \in R]$. □

4.3 Proof of Theorem 3.2(a)–(d)

(a) If the loop is entered at x, then $x \in E_+ \subseteq R_X$. Apply Lemma 4.2(ii).

(c) The first line returns x with probability $p_X(x) \min\{1, p_Y(x)/p_X(x)\} = \min\{p_X(x), p_Y(x)\}$. The loop is entered only at $x \in E_+$ and returns only some w with $\rho(w) < 1$, that is, $w \in E_-$. Since $E_+ \cap E_- = \emptyset$, the loop never returns x. Hence $\hat{P}(x,x) = \min\{p_X(x), p_Y(x)\}$, and the off-diagonal support lies in $E_+ \times E_-$.

(b) By (a), $\hat{P}(x, E) = p_X(x)$. For the Y-marginal, let $y \in E_-$. By Lemma 4.1(ii), $y \in R$, so $P(R,y) = p_Y(y)$. Using Lemma 4.2(iv), the expected number of exits at y is

$$\hat{P}(E \setminus \{y\}, y) = \sum_v \mu(v) H(v,y) = \sum_v T(v,y)(1 - \rho(y)) = P(R,y)\Big(1 - \frac{p_X(y)}{p_Y(y)}\Big) = \beta(y).$$

For $y \notin E_-$ there are no exits at y, and again $\hat{P}(E \setminus \{y\}, y) = 0 = \beta(y)$. With (c), $\hat{P}(E, y) = \min\{p_X(y), p_Y(y)\} + \beta(y) = p_Y(y)$.

(d) Let $q_x(y) = \mathbb{P}[\text{MRC-Cond}_P(x) = y]$. Number the invocations $k = 1, 2, \ldots$, let I_k be the input of the k-th one (undefined if there are fewer than k), and let $D_k(v)$ be the number of conditional-sampler calls with input v that it makes. The k-th invocation uses fresh randomness, so $\mathbb{E}[D_k(v) \mid I_k = x, \text{past}] = N_x(v)$. By the tower property and Tonelli's theorem,

$$\mathbb{E}\Big[\sum_k D_k(v)\Big] = \sum_x c(x) N_x(v) \le \lambda \sum_x p_X(x) N_x(v) = \lambda\, \mu(v) \le \lambda\, p_X(v).$$

The same argument gives $\sum_x c(x) q_x(y) \le \lambda \sum_x p_X(x) q_x(y) = \lambda\, \hat{P}(E, y) = \lambda\, p_Y(y)$ for the expected number of invocations returning y. One run of MRC has $c = p_X$ and $\lambda = 1$, and makes $\sum_v \mu(v) = p_X(R) \le 1$ expected conditional-sampler calls. □

4.4 Trajectories, and proof of Theorem 3.2(e),(f)

In one run of MRC, if the loop is entered, let $t = (v_0, v_1, \ldots, v_l)$ be its *trajectory*: $v_0 = x$, $v_1, \ldots, v_{l-1}$ are the values at which it continued, v_l is the returned value, and $l \ge 1$. If the loop is not entered, let $t = \emptyset$. Writing e_{vw} for the matrix unit, define

$$g(t) = \sum_{i=1}^{l} e_{v_{i-1} v_i}, \qquad \hat{g}(t) = \sum_{i=1}^{l-1} e_{v_i v_i} + e_{v_0 v_l}, \qquad g(\emptyset) = \hat{g}(\emptyset) = 0.$$

So $g(t)$ records the arcs the walk used. $\hat{g}(t)$ replaces the walk by a single shortcut arc from its start to its end, plus a self-loop at every interior visit. Let P_U be the restriction of P to $U \times U$ (zero elsewhere), and $D_U = \operatorname{diag}(p_X(u)\mathbf{1}[u \in U])$.

Lemma 4.3 (trajectory decomposition)**.** $\mathbb{E}[g(t)] = P - P_U$ *and* $\mathbb{E}[\hat{g}(t)] = \hat{P} - D_U$.

Proof. $\mathbb{E}[g(t)] = T$ by Lemma 4.2(iv). By Lemma 4.1(i), $P(v, w) > 0$ with $v \in U$ forces $w \in U$. So $T = P - P_U$.

For $\hat{g}$, off-diagonal entries come only from the shortcut arc (note $v_0 \ne v_l$), and $\mathbb{E}[\hat{g}(t)]_{xy}$ for $x \ne y$ is the probability that the run starts at x and the loop returns y, which is $\hat{P}(x, y)$. A diagonal entry (w, w) counts continuations at w. Its expectation is $\sum_v \mu(v) Q(v, w) = \sum_v T(v, w) \rho(w) = P(R, w) \rho(w)$. For $w \in R$ this equals $p_Y(w)\rho(w) = \min\{p_X(w), p_Y(w)\} = \hat{P}(w, w)$. For $w \in U$ it is $0 = \hat{P}(w, w) - p_X(w)$, because $p_X(w) = p_Y(w)$ there. □

In words, apart from a balanced component on U, which the walk never visits and which MRC replaces by its diagonal coupling, the old coupling is the average of the walks and the new coupling is the average of the shortcut walks.

Lemma 4.4 (pathwise inequalities)**.** *For every trajectory t, every $S \subseteq E$ and every partition $\mathcal{F}$ of E:*

(P1) $g(t)(S, E \setminus S) \ge \hat{g}(t)(S, E \setminus S)$*;*

(P2) $\sum_{S \ne T \in \mathcal{F}} (g - \hat{g})(t)(S, T) \ge \sum_{S \ne T \in \mathcal{F}} ((\hat{g} - g)(t)(S, T))^+$.

Proof. (P1) The self-loops of $\hat{g}(t)$ do not cross from S to $E \setminus S$, so the right side is 1 if $v_0 \in S$ and $v_l \notin S$, and 0 otherwise. In the first case the walk leaves S at some step, so the left side is at least 1.

(P2) Let a be the number of steps of t that cross between different blocks of $\mathcal{F}$. Then $\sum_{S \neq T} g(t)(S,T) = a$. The only off-block entry of $\hat{g}(t)$ is the shortcut, so $\sum_{S \neq T} \hat{g}(t)(S,T) = b \in \{0,1\}$, with $b = 1$ iff v_0 and v_l lie in different blocks $S_0 \neq T_0$. The right side is 1 iff $b = 1$ and no step of t goes directly from S_0 to T_0, and 0 otherwise. If it is 1, the walk leaves S_0 first to a block other than T_0, so it crosses at least twice, and $a - b \geq 1$. If it is 0, then $a - b \geq 0$, because $b = 1$ implies $a \geq 1$. □

Proof of Theorem 3.2(e). Rows of P and $\hat{P}$ both sum to p_X, so $\hat{P}(S,S) - P(S,S) = P(S, E \setminus S) - \hat{P}(S, E \setminus S)$. By Lemma 4.3, this equals $\mathbb{E}[(g - \hat{g})(t)(S, E \setminus S)] + P_U(S, E \setminus S)$, since D_U is diagonal. Both terms are nonnegative, the first by (P1). □

Proof of Theorem 3.2(f). For distinct blocks S, T, $D_U(S,T) = 0$, so by Lemma 4.3 $(P - \hat{P})(S,T) = \mathbb{E}[(g - \hat{g})(t)(S,T)] + P_U(S,T)$. Summing, using (P2) and then the convexity of $z \mapsto z^+$,

$$\begin{aligned} \sum_{S \neq T} (P - \hat{P})(S,T) &\geq \sum_{S \neq T} \Big((\mathbb{E}[(\hat{g} - g)(t)(S,T)])^+ + P_U(S,T) \Big) \\ &\geq \sum_{S \neq T} \left(\mathbb{E}[(\hat{g} - g)(t)(S,T)] - P_U(S,T) \right)^+ . \end{aligned}$$

The last step uses $z^+ + u \geq (z - u)^+$ for $u \geq 0$. The argument of the last positive part is $(\hat{P} - P)(S,T)$. □

Proof of Corollary 3.3. Let $\mathcal{F}$ be the partition of E into fibres of f, so $\gamma_f = \sum_{S \in \mathcal{F}} (\hat{P} - P)(S,S)$. Part (i) is Theorem 3.2(e). For (ii), by (e) the diagonal blocks contribute exactly γ_f to $d_{\mathrm{TV}}(P^f, \hat{P}^f) = \sum_{S,T} \Big((\hat{P} - P)(S,T) \Big)^+$. By (f), the off-diagonal blocks contribute at most $\sum_{S \neq T} (P - \hat{P})(S,T)$, which is γ_f because P and $\hat{P}$ have the same total mass. For (iii), $\hat{P}^f$ is a coupling of the same pushforward marginals as P^f by Theorem 3.2(b). Since P^f maximizes the agreement probability among such couplings, $\gamma_f \leq 0$. With (i), $\gamma_f = 0$, and then (ii) gives $\hat{P}^f = P^f$. □

4.5 Proof of the closed form

Proof of Proposition 3.4. The diagonal and the support were established in Theorem 3.2(c). Fix $x \in E_+$ and $y \in E_-$. Summing over the number of continuations,

$$\hat{P}(x,y) = \alpha(x) \sum_{n \geq 0} \sum_{v} (Q^n)_{xv} H(v,y).$$

On E_X, using $\rho(w) = p_X(w)/p_{\max}(w)$, we have $Q = D_X^{-1} P D_{\max}^{-1} D_X$, where $D_X = \mathrm{diag}(p_X)$ and $D_{\max} = \mathrm{diag}(p_{\max})$. Also $H(v,y) = P(v,y)\beta(y)/(p_X(v)p_{\max}(y))$. Hence

$$\sum_{v} (Q^n)_{xv} H(v,y) = \frac{\beta(y)}{p_X(x)p_{\max}(y)} \left[(P D_{\max}^{-1})^n P \right]_{xy} = \frac{\beta(y)}{p_X(x)p_{\max}(y)} \sqrt{p_{\max}(x)p_{\max}(y)}\, (A^{n+1})_{xy},$$

since $(PD_{\max}^{-1})^n P = D_{\max}^{1/2} A^{n+1} D_{\max}^{1/2}$ on the support of $p_{\max}$. Intermediate nodes of a positive-weight path have $p_{\max} > 0$, so restricting to that support loses nothing. Using $p_{\max}(x) = p_X(x)$ and $p_{\max}(y) = p_Y(y)$ gives the series form.

For convergence, and for the pseudoinverse, order the nodes as R followed by U. By Lemma 4.1(i), $A = A_R \oplus A_U$ is block diagonal. Ordering R as R_X followed by $R \setminus E_X$ (rows of A vanish on the latter) makes A_R block upper triangular, with diagonal blocks A_{R_X} and 0. Now A_{R_X} is similar to Q_R, since $Q = D_X^{-1} D_{\max}^{1/2} A D_{\max}^{-1/2} D_X$ on E_X. So $\varrho(A_R) = \varrho(Q_R) < 1$ by Lemma 4.2(ii). Hence $\sum_{n\geq 1}(A^n)_{xy} = \sum_{n\geq 1}(A_R^n)_{xy} = [(I - A_R)^{-1}]_{xy}$, using $x \neq y$ to add the $n = 0$ term. The Moore–Penrose pseudoinverse of a block diagonal matrix is the block diagonal matrix of the blocks' pseudoinverses. Permutation similarity commutes with taking pseudoinverses, and $(I - A_R)^+ = (I - A_R)^{-1}$. Therefore $[(I - A)^+]_{xy} = [(I - A_R)^{-1}]_{xy}$. Finally, a node of U cannot reach a sink, because U is closed and contains no sinks. So under the extra hypothesis every node of U has $p_{\max} = 0$, $A_U = 0$, and $I - A = (I - A_R) \oplus I$ is invertible. □

Remark 4.5 (path form)**.** Equivalently, for $x \in E_+$ and $y \in E_-$,

$$\hat{P}(x, y) = \alpha(x)\beta(y) \sum_{l\geq 1} \sum_{v_0 = x,\, v_1, \ldots, v_{l-1},\, v_l = y} \frac{\prod_{i=1}^{l} P(v_{i-1}, v_i)}{\prod_{i=0}^{l} p_{\max}(v_i)},$$

where the inner sum runs over positive-weight paths, so every $p_{\max}(v_i) > 0$. This is the probability of each trajectory, summed. The summand is visibly invariant under reversing the path and transposing P.

5 Simultaneously maximal coupling on chain-product trees

From now on, $X = (X_i)_{i\in V}$ and $Y = (Y_i)_{i\in V}$, where V is a nonempty finite index set of size n and X_i, Y_i take values in a finite set E_i. For $S \subseteq V$ write $E_S = \prod_{i\in S} E_i$, and write x_S for the restriction of x to S. Let P_X and P_Y be the laws of X and Y on E_V.

Definition 5.1 (simultaneously maximal coupling)**.** Let $\mathcal{F}$ be a family of nonempty subsets of V. A coupling of P_X and P_Y is *simultaneously maximal for* $\mathcal{F}$ if, for every $S \in \mathcal{F}$, (X_S, Y_S) is maximally coupled: $\mathbb{P}[X_S = Y_S = a] = \min\{\mathbb{P}[X_S = a], \mathbb{P}[Y_S = a]\}$ for all $a \in E_S$. $\mathcal{F}$ is a *chain* if it is totally ordered by inclusion, and *laminar* if any two members are nested or disjoint.

Definition 5.2 (chain-product tree)**.** A *chain-product tree* (CPT) over V is a rooted tree whose nodes τ have *scopes* $V(\tau) \subseteq V$, built from three kinds of node:

- a *leaf* $\mathrm{leaf}(i)$, with scope $\{i\}$;
- a *product* $\mathrm{prod}(\tau_1, \tau_2)$ with $V(\tau_1) \cap V(\tau_2) = \emptyset$, with scope $V(\tau_1) \cup V(\tau_2)$;
- a *chain* $\mathrm{chain}(\tau_1, i)$ with $i \notin V(\tau_1)$, with scope $V(\tau_1) \cup \{i\}$.

The root has scope V, so each $i \in V$ is introduced exactly once, by a leaf or by a chain node. A pair (P_X, P_Y) is *compatible* with the CPT if, for every product node $\mathrm{prod}(\tau_1, \tau_2)$, $X_{V(\tau_1)}$ and $X_{V(\tau_2)}$ are independent under P_X, and likewise for Y under P_Y.

A compatible distribution factorizes along the tree. The marginal of $X_{V(\tau)}$ is P_{X_i} at a leaf, the product of the children's marginals at a product node, and the child's marginal times the

conditional $P(X_i \mid X_{V(\tau_1)})$ at a chain node. The *oracles* of a CPT are, for $Z \in \{X, Y\}$: a sampler and likelihood for P_{Z_i} at each leaf, and a conditional sampler and conditional likelihood for $P(Z_i \mid Z_{V(\tau_1)} = z)$ at each chain node. The conditional oracles must be defined for every z, including histories of probability zero, where any fixed normalized distribution may be used. Since such histories have zero marginal likelihood, the choice never affects a likelihood. It also never affects a coupling, and the extended kernels are used in the definition of $\ltimes$ below. Implementations may short-circuit a likelihood to 0 as soon as a factor vanishes. Likelihoods of $Z_{V(\tau)}$ are computed as products of oracle likelihoods, at a cost of $|V(\tau)|$ oracle evaluations. Ancestral sampling along the tree draws $x \sim P_X$ with one call to each X-side sampler.

Couplings at the nodes. Define a coupling $\hat{P}_\tau$ of $(P_{X_{V(\tau)}}, P_{Y_{V(\tau)}})$ for each node, bottom-up:

$$\begin{aligned}
\hat{P}_{\text{leaf}(i)} &= \text{MRC}(P_{X_i} \otimes P_{Y_i}), \\
\hat{P}_{\text{prod}(\tau_1,\tau_2)} &= \text{MRC}\big(\hat{P}_{\tau_1} \otimes \hat{P}_{\tau_2}\big), \\
\hat{P}_{\text{chain}(\tau_1,i)} &= \text{MRC}\big(\hat{P}_{\tau_1} \ltimes (K_X, K_Y)\big),
\end{aligned}$$

where $(\hat{P}_{\tau_1} \otimes \hat{P}_{\tau_2})((x_1, x_2), (y_1, y_2)) = \hat{P}_{\tau_1}(x_1, y_1)\hat{P}_{\tau_2}(x_2, y_2)$, and

$$(J \ltimes (K_X, K_Y))((x_1, x_i), (y_1, y_i)) = J(x_1, y_1)\, K_X(x_i \mid x_1)\, K_Y(y_i \mid y_1),$$

with $K_Z(\cdot \mid z) = P(Z_i = \cdot \mid Z_{V(\tau_1)} = z)$. In words: the children's couplings are combined independently (product), or extended by letting X_i and Y_i follow their own conditionals given their own pasts (chain). The result is then recoupled.

The SMC-CPT algorithm. Every coupling above has a conditional sampler built from those of its parts:

- $\text{MRC}(J)$: run MRC-Cond$_J$ (Algorithm 2) with J's conditional sampler and the likelihoods of the scope's marginals;
- $P_{X_i} \otimes P_{Y_i}$: ignore the input and draw from P_{Y_i};
- $J_1 \otimes J_2$: given (x_1, x_2), return (y_1, y_2) with independent $y_1 \sim J_1(\cdot \mid x_1)$, $y_2 \sim J_2(\cdot \mid x_2)$;
- $J \ltimes (K_X, K_Y)$: given (x_1, x_i), draw $y_1 \sim J(\cdot \mid x_1)$, then $y_i \sim K_Y(\cdot \mid y_1)$.

SMC-CPT draws $x \sim P_X$ by ancestral sampling and returns (x, y) with y from the root's conditional sampler. Appendix B spells this out as pseudocode.

Theorem 5.3 (SMC-CPT)**.** *Let (P_X, P_Y) be compatible with a CPT. Then:*

(a) ***Validity.*** *SMC-CPT terminates almost surely, and its output has law $\hat{P}_{\text{root}}$, a coupling of P_X and P_Y.*

(b) ***Simultaneous maximality.*** *For every node τ, $(X_{V(\tau)}, Y_{V(\tau)})$ is maximally coupled. Moreover the joint law of $(X_{V(\tau)}, Y_{V(\tau)})$ under $\hat{P}_{\text{root}}$ equals $\hat{P}_\tau$.*

(c) ***Efficiency.*** *In one run, each X-side sampler is called exactly once. At a leaf, the expected number of draws from P_{Y_i} is at most 1. At a chain node, the expected number of draws from $P(Y_i \mid Y_{V(\tau_1)} = z)$ is at most $\mathbb{P}[Y_{V(\tau_1)} = z]$, so at most 1 in total. At each node, the recoupling step is invoked at most once in expectation and runs at most one loop iteration in expectation. So the expected number of oracle likelihood evaluations is at most $4\sum_\tau |V(\tau)| \le 4n(h+1)$, where h is the height of the tree.*

Proof. Couplings and maximality, bottom-up. We claim that $\hat{P}_\tau$ is a coupling of the marginals on $V(\tau)$, and that for every descendant σ of τ (including τ) the joint law of $(X_{V(\sigma)}, Y_{V(\sigma)})$ under $\hat{P}_\tau$ is $\hat{P}_\sigma$, which is maximal.

At a leaf this is Theorem 3.2(b),(c).

At a product node, $\hat{P}_{\tau_1} \otimes \hat{P}_{\tau_2}$ has X-marginal $P_{X_{V(\tau_1)}} \otimes P_{X_{V(\tau_2)}} = P_{X_{V(\tau)}}$ by compatibility, and likewise for Y. Its projection onto the coordinates of a descendant σ of τ_j is $\hat{P}_\sigma$ by induction, and $\hat{P}_\sigma$ is maximal. Corollary 3.3(iii), with f the projection onto $E_{V(\sigma)}$, says that MRC preserves each of these projections. Theorem 3.2 makes the result a maximal coupling of the marginals on $V(\tau)$.

At a chain node, $\hat{P}_{\tau_1} \ltimes (K_X, K_Y)$ has X-marginal $P_{X_{V(\tau_1)}} K_X = P_{X_{V(\tau)}}$, and likewise for Y. Its projection onto $V(\tau_1)$ is $\hat{P}_{\tau_1}$, and we conclude as before.

Sampling. By induction, each conditional sampler draws from the correct conditional law. For MRC(J) this is Theorem 3.2 applied to J. For the product and chain constructions it is immediate from the definitions, since, for example, $(J \ltimes K)(\cdot \mid x_1, x_i)$ is the law of (y_1, y_i) with $y_1 \sim J(\cdot \mid x_1)$ and $y_i \sim K_Y(\cdot \mid y_1)$. Termination follows from the efficiency bounds below: every procedure makes a finite expected number of oracle calls, hence finitely many almost surely, and the loop of each MRC-Cond exits almost surely by Theorem 3.2(a).

Efficiency, top-down. For a node τ let M_τ be the conditional sampler of $\hat{P}_\tau$ and B_τ that of the coupling before recoupling. We show that for every $x \in E_{V(\tau)}$, the expected numbers of invocations of M_τ and of B_τ with input x are both at most $\mathbb{P}[X_{V(\tau)} = x]$. At the root, M is invoked once, with $x \sim P_X$. From M_τ to B_τ, apply Theorem 3.2(d) with $\lambda = 1$. From B_τ to the children: at a product node, each invocation of B_τ with input (x_1, x_2) makes one invocation of M_{τ_1} with input x_1, so the count at x_1 is at most $\sum_{x_2} \mathbb{P}[X_{V(\tau)} = (x_1, x_2)] = \mathbb{P}[X_{V(\tau_1)} = x_1]$, and likewise for τ_2. At a chain node, each invocation makes one invocation of M_{τ_1} with input x_1, bounded the same way, followed by one draw from $K_Y(\cdot \mid y_1)$, where y_1 is that invocation's output. By the second half of Theorem 3.2(d), the expected number of invocations of M_{τ_1} that return y_1 is at most $\mathbb{P}[Y_{V(\tau_1)} = y_1]$, which bounds the draws from $K_Y(\cdot \mid y_1)$. At a leaf, every invocation of B_τ makes one draw from P_{Y_i}, giving at most $\sum_x \mathbb{P}[X_i = x] = 1$ draws. Finally, M_τ is invoked at most once in expectation, and its loop iterations are invocations of B_τ, also at most one in expectation. Each invocation and each iteration evaluates the two likelihoods of a point of $E_{V(\tau)}$ once, which costs at most $2 \cdot 2|V(\tau)|$ oracle evaluations per node. Summing, $\sum_\tau |V(\tau)| = \sum_i$(number of nodes whose scope contains i) $\le n(h+1)$. □

Remark 5.4 (what the nodes buy)**.** Theorem 5.3 guarantees simultaneous maximality on the scopes of all nodes. For particular distributions, other subsets may happen to be maximal as well, for example every subset when $P_X = P_Y$. The guaranteed scopes form a laminar family. A laminar family of nonempty subsets of an n-set has at most $2n - 1$ members, and a CPT built entirely of leaves and products realizes all $2n - 1$.

6 Characterizations

In this section every alphabet E_i has at least two elements. For the counterexamples we fix two of them, call them 0 and 1, and set every unmentioned coordinate to the constant 0.

Lemma 6.1 (CPTs from laminar families)**.** *Let $\mathcal{F}$ be laminar. For $S \in \mathcal{F} \cup \{V\}$, its* children *are the maximal members of $\mathcal{F}$ strictly contained in S. These are pairwise disjoint. Suppose that for every $S \in \mathcal{F} \cup \{V\}$ with children $C_1, \dots, C_k$, the variables $X_{C_1}, \dots, X_{C_k}$ are mutually independent, and likewise $Y_{C_1}, \dots, Y_{C_k}$. Then there is a CPT, compatible with (P_X, P_Y), in which every member of $\mathcal{F}$ is the scope of a node. Hence (P_X, P_Y) admits a coupling that is simultaneously maximal for $\mathcal{F}$, and SMC-CPT samples one.*

Proof. Build a node τ_S with scope S recursively over the inclusion forest. Let $C_1, \dots, C_k$ be the children of S and $S \setminus \bigcup_j C_j = \{r_1, \dots, r_m\}$ in any order. If $k \geq 1$, take $\mathrm{prod}(\cdots \mathrm{prod}(\tau_{C_1}, \tau_{C_2}) \cdots, \tau_{C_k})$ and then chain in $r_1, \dots, r_m$. If $k = 0$, then $m \geq 1$: take $\mathrm{leaf}(r_1)$ and chain in $r_2, \dots, r_m$. Compatibility at the product nodes is mutual independence of the children. Apply the construction to $S = V$ to get the root, then apply Theorem 5.3. □

Theorem 6.2 (no structural assumptions)**.** *Every pair (P_X, P_Y) of distributions on E_V admits a coupling that is simultaneously maximal for $\mathcal{F}$ iff $\mathcal{F}$ is a chain.*

Proof. If $\mathcal{F}$ is a chain, each $S \in \mathcal{F}$ has at most one child, and Lemma 6.1 applies with no independence needed. The resulting CPT is a single chain of chain nodes. Conversely, suppose $S, T \in \mathcal{F}$ are incomparable, and take $i \in S \setminus T$ and $j \in T \setminus S$. Let $X_i = X_j \sim \mathrm{Unif}\{0,1\}$, and let $Y_i \sim \mathrm{Unif}\{0,1\}$ with $Y_j = 1 - Y_i$. Then X_S and Y_S have the same law, so a maximal coupling of them has $X_S = Y_S$, and in particular $X_i = Y_i$, almost surely. Similarly $X_j = Y_j$ almost surely. But then $Y_i = X_i = X_j = Y_j$, contradicting $Y_j = 1 - Y_i$. □

Theorem 6.3 (independent coordinates)**.** *Every pair (P_X, P_Y) of product distributions (the X_i mutually independent, and the Y_i mutually independent) admits a coupling that is simultaneously maximal for $\mathcal{F}$ iff $\mathcal{F}$ is laminar.*

Proof. If $\mathcal{F}$ is laminar, disjoint sets of coordinates are independent, so Lemma 6.1 applies. Conversely, suppose $S, T \in \mathcal{F}$ cross: take $i \in S \cap T$, $j \in S \setminus T$ and $k \in T \setminus S$. Let $X_i = 1$, and let X_j, X_k be independent and uniform on $\{0,1\}$. Let Y_i be uniform on $\{0,1\}$ and $Y_j = Y_k = 1$. Then X_S is uniform on two points, which differ in coordinate j, and Y_S is uniform on two points, which differ in coordinate i. The only common point is the one with $x_i = x_j = 1$. A maximal coupling of (X_S, Y_S) therefore puts mass $1/2$ on $X_S = Y_S$, and that event is contained in both $\{X_j = 1\}$ and $\{Y_i = 1\}$, which have probability $1/2$ each. Hence $\{X_j = 1\} = \{Y_i = 1\}$ almost surely. By the same argument for T, $\{X_k = 1\} = \{Y_i = 1\}$ almost surely. So $X_j = X_k$ almost surely, contradicting their independence. □

Remark 6.4. Theorem 6.3 answers an open question from the notes. With independent tokens, the whole response and every single position can be maximally coupled at once, since $\{\{1\}, \dots, \{n\}, V\}$ is laminar.

For Bayesian networks, let G be a DAG on V. A distribution on E_V is *Markov with respect to G* if it factorizes as $\prod_v P(x_v \mid x_{\mathrm{pa}(v)})$. Let $\mathrm{An}(S)$ be the set of ancestors of members of S, including S itself, and call S *ancestral* if $\mathrm{An}(S) = S$. Two sets S, T are d-separated by $\emptyset$ iff no active path joins them, that is, iff $\mathrm{An}(S) \cap \mathrm{An}(T) = \emptyset$: an active path without conditioning has no collider, so it is a *trek* through a common ancestor.

Theorem 6.5 (Bayesian networks)**.** *Let G be a DAG on V. Every pair (P_X, P_Y) of distributions that are Markov with respect to G admits a coupling that is simultaneously maximal for $\mathcal{F}$ iff $\mathcal{F}$ is laminar and $\mathrm{An}(S) \cap \mathrm{An}(T) = \emptyset$ for all disjoint $S, T \in \mathcal{F}$.*

Proof. If. We check the hypothesis of Lemma 6.1. The children $C_1, \ldots, C_k$ of any $S \in \mathcal{F} \cup \{V\}$ are pairwise disjoint members of $\mathcal{F}$, so the ancestral sets $A_j = \mathrm{An}(C_j)$ are pairwise disjoint. No arc joins two of them, since an arc $u \to v$ with $v \in A_j$ puts $u \in A_j$. The marginal of a G-Markov distribution on the ancestral set $A = \bigcup_j A_j$ is $\prod_{v \in A} P(x_v \mid x_{\mathrm{pa}(v)})$. This splits as a product of factors, the j-th depending only on x_{A_j} and being the marginal of X_{A_j}. So $X_{A_1}, \ldots, X_{A_k}$ are mutually independent, and hence so are $X_{C_1}, \ldots, X_{C_k}$. The same holds for Y.

Only if, crossing. If $\mathcal{F}$ is not laminar, the product distributions of Theorem 6.3 are Markov with respect to every DAG.

Only if, common ancestor. Suppose $S, T \in \mathcal{F}$ are disjoint and $\mathrm{An}(S) \cap \mathrm{An}(T) \neq \emptyset$. A *trek* between s and t is a pair of directed paths π_s from a vertex c to s and π_t from c to t, either possibly of length 0, that share only c. If $c \in \mathrm{An}(s) \cap \mathrm{An}(t)$, take directed paths from c to s and t and restart both at the last vertex of the first path that lies on the second. This gives a trek. Choose $s \in S$, $t \in T$ and a trek between them of minimum total length. Then s is its only vertex in S and t its only vertex in T. Indeed, any sub-path of a trek is again a trek (it has no collider). So if another trek vertex u were in T, the part of the trek between s and u would be a strictly shorter trek between S and T. If $u \in S$, the part between u and t would be. Since $s \neq t$, at least one of π_s, π_t has positive length. Say π_t does, swapping the roles of S and T if needed.

Now define X and Y. Let $X_c, Y_c \sim \mathrm{Unif}\{0, 1\}$. Every other vertex u on the trek copies its predecessor u^- on its path, which is a parent of u: $X_u = X_{u^-}$ and $Y_u = Y_{u^-}$, except that $Y_t = 1 - Y_{t^-}$. All other vertices are constant. Each conditional depends on at most one parent, so both laws are G-Markov. Now $X_s = X_t$, $Y_s = 1 - Y_t$, and each of X_s, X_t, Y_s, Y_t is uniform. Since S contains no trek vertex other than s, X_S and Y_S have the same law, so maximality forces $X_s = Y_s$ almost surely. Likewise $X_t = Y_t$ almost surely. Then $Y_s = X_s = X_t = Y_t$, which is a contradiction. □

Corollary 6.6 (ancestral laminar families: an efficient algorithm)**.** *Let $\mathcal{F}$ be laminar with every member ancestral in G. Then the CPT of Lemma 6.1, with each remainder $S \setminus \bigcup_j C_j$ chained in a topological order of G, has chain-node conditionals $P(Z_r \mid Z_{\mathrm{prev}}) = P(Z_r \mid Z_{\mathrm{pa}(r)})$ for $Z \in \{X, Y\}$. These are the network's own conditional distributions. The identity holds on histories of positive probability, and elsewhere the network's conditional distributions serve as the extension required of the oracles. So SMC-CPT runs with oracle access only to the conditional distributions of the two networks, with the costs of Theorem 5.3(c).*

Proof. Disjoint ancestral sets have disjoint ancestor sets, so Theorem 6.5 applies. Let prev be the set of vertices introduced before r in the construction of S. The parents of r lie in S, because S is ancestral, and each lies either in a child C_j or earlier in the remainder, so $\mathrm{pa}(r) \subseteq \mathrm{prev}$. No vertex of prev is a descendant of r: the C_j are ancestral and do not contain r, and the earlier remainder vertices precede r topologically. The local Markov property gives $X_r \perp X_{\mathrm{prev} \setminus \mathrm{pa}(r)} \mid X_{\mathrm{pa}(r)}$. Every scope in this CPT is ancestral, so its marginal likelihood is a product of network conditionals. □

Remark 6.7 (non-ancestral families)**.** When $\mathcal{F}$ satisfies the condition of Theorem 6.5 but is not ancestral, SMC-CPT still applies, but its chain nodes need conditionals such as $P(X_r \mid X_{\mathrm{prev}})$ given non-ancestral sets. These generally require marginalization and are not available from the network's own conditionals.

Example 6.8 (star graphs)**.** Theorem 6.5 can be counterintuitive. Let G be a star with center 0 and leaves $1, \dots, n-1$. If the arcs point *inward* (the leaves are independent causes of the center), then any laminar family of leaf sets (including the leaf singletons) is allowed, plus V, for up to $2(n-1)-1+1 = 2n-2$ sets ($n \geq 2$). The center singleton is not allowed alongside any leaf set, because $\mathrm{An}(\{0\}) = V$. If the arcs point *outward*, then any two disjoint nonempty sets share the ancestor 0, so $\mathcal{F}$ must be a chain. That allows at most n sets, no better than with no assumptions, even though the leaves are conditionally independent given the center. An explicit witness: $X_0, Y_0 \sim \mathrm{Unif}(\mathbb{Z}_n)$, $X_i = X_0$ and $Y_i = Y_0 + i \bmod n$ for $1 \leq i \leq n-1$. Every pair (X_i, Y_i) is identically distributed and could be coupled perfectly on its own, yet $X_i = Y_i$ holds for exactly one index $i \in \{0, \dots, n-1\}$.

Remark 6.9 (discrete diffusion)**.** In discrete-diffusion samplers such as the absorbing-state dynamics of SEDD [11], the per-position decisions of one step are conditionally independent given the current state. Suppose we model the sampling decisions as a Bayesian network in which every decision of step $t \geq 2$ has all decisions of earlier steps as parents, which is the generic situation for a neural denoiser. Then the decisions of the first step are independent. Every set containing a later decision has every first-step decision as an ancestor, so by Theorem 6.5 such a set must be nested with every other member of $\mathcal{F}$. The guarantees that hold for *all* distributions with this dependency graph are therefore a laminar family on the first step, plus a chain of sets above it, as in the outward star of Example 6.8. Particular models may of course have more independence than this graph records.

7 The autoregressive case

Now take $V = \{1, \dots, n\}$ with no independence assumed, which is the setting of autoregressive generation. Let $\mathcal{F}$ be the chain of prefixes $\{1..i\}$. The CPT is leaf(1) followed by chain nodes for $2, \dots, n$. Write $x_{\leq i}$ for prefixes, and write $p_X(x_{\leq i})$ and $p_Y(x_{\leq i})$ for the prefix marginals.

Definition 7.1 (APOC)**.** The *all-prefix-optimal coupling* Π_n is defined recursively. Π_1 is the CMC coupling of X_1 and Y_1. Given Π_{i-1}, for $a \in E_{\leq i-1}$ let $r_X(z \mid a) \propto (p_X(az) - p_Y(az))^+$ and $r_Y(z \mid a) \propto (p_Y(az) - p_X(az))^+$ be normalized residuals, and let $m(a) = \Pi_{i-1}(a, a) - \sum_z \min\{p_X(az), p_Y(az)\}$ be the probability that the prefixes agree on a but the next tokens disagree. Then

$$\Pi_i(x_{\leq i}, y_{\leq i}) = \begin{cases} \min\{p_X(x_{\leq i}), p_Y(x_{\leq i})\} & x_{\leq i} = y_{\leq i}, \\ m(x_{<i})\, r_X(x_i \mid x_{<i})\, r_Y(y_i \mid x_{<i}) & x_{<i} = y_{<i},\ x_i \neq y_i, \\ \Pi_{i-1}(x_{<i}, y_{<i})\, r_X(x_i \mid x_{<i})\, r_Y(y_i \mid y_{<i}) & x_{<i} \neq y_{<i}. \end{cases}$$

Terms with zero mass are 0, and the normalizers are positive whenever the mass is.

Π_n is maximal for every prefix. Up to notation, this is the coupling the author called APOC. The author noted that the first version of Sun et al. [15] (BlockVerify, later called greedy block verification) is essentially its conditional sampler. All-prefix-maximal couplings of processes are the maximal agreement couplings of [16, 3].

Theorem 7.2 (SMC-CPT reproduces APOC)**.** *For the prefix CPT,* $\hat{P}_{\mathrm{root}} = \Pi_n$.

Proof. By induction on i. At the leaf, MRC of the independent coupling is CMC (Remark 3.1). For the step, let $J = \Pi_{i-1}$, which is maximal on $E_1 := E_{\le i-1}$, let $P = J \ltimes (K_X, K_Y)$ on $E = E_1 \times E_i$, and let $\hat{P} = \mathrm{MRC}(P)$. Write $x = (x_1, x_2)$ with $x_1 \in E_1$ and $x_2 \in E_i$.

Step 1: the walk only sees the first block. For $v = (v_1, v_2)$, $P(\cdot \mid v)$ is the law of (w_1, w_2) with $w_1 \sim J(\cdot \mid v_1)$ and $w_2 \sim K_Y(\cdot \mid w_1)$. This does not depend on v_2. So the law of the loop started at x depends only on x_1, and $\hat{P}(x, y) = \alpha(x)h(x_1, y)$ for $x \neq y$, for some function h.

Step 2: symmetry. $P^\top = J^\top \ltimes (K_Y, K_X)$ has the same form with the roles of X and Y exchanged. By Corollary 3.5 and Step 1 applied to $P^\top$, $\hat{P}(x, y) = \beta(y)h'(y_1, x)$ for $x \neq y$.

Step 3: product form. Fix x_1, y_1. On $\{x_2 : \alpha(x) > 0\} \times \{y_2 : \beta(y) > 0\}$, the quantity $h(x_1, y)/\beta(y) = h'(y_1, x)/\alpha(x)$ depends on y_2 alone on the left and on x_2 alone on the right, so it is a constant $C(x_1, y_1)$. Outside this set, $\hat{P}(x, y) = 0$ by Theorem 3.2(c). Note that $x = y$ cannot have $\alpha(x)\beta(y) > 0$. Hence $\hat{P}(x, y) = C(x_1, y_1)\alpha(x)\beta(y)$ for all $x \neq y$.

Step 4: identify C. The projection of P onto E_1 is J, which is maximal, so by Corollary 3.3(iii) $\hat{P}$ has the same projection. For $x_1 \neq y_1$ this gives $J(x_1, y_1) = C(x_1, y_1)\, a(x_1)\, b(y_1)$, where $a(x_1) = \sum_{x_2} \alpha(x_1, x_2)$ and $b(y_1) = \sum_{y_2} \beta(y_1, y_2)$. If $J(x_1, y_1) > 0$, then x_1 is a source of J (Lemma 2.2), so $a(x_1) \ge p_X(x_1) - p_Y(x_1) > 0$, and likewise $b(y_1) > 0$. For $x_1 = y_1$, the off-diagonal mass inside the block is $J(x_1, x_1) - \sum_z \min\{p_X(x_1 z), p_Y(x_1 z)\} = C(x_1, x_1)a(x_1)b(x_1)$. Since $\alpha(x)/a(x_1) = r_X(x_2 \mid x_1)$ and $\beta(y)/b(y_1) = r_Y(y_2 \mid y_1)$, these identities are exactly the recursion defining Π_i. □

7.1 Per-token couplings

Two simpler couplings of autoregressive models are common in practice. *Per-token optimal coupling* (PTOC) maximally couples the next-token distributions given the current prefixes. This is the coupling implicit in standard speculative decoding [8, 2]. *Per-token common random numbers* (PTCRN) is a model of seed reuse: $X_i = f_X(U_i, X_{<i})$ and $Y_i = f_Y(U_i, Y_{<i})$ with shared i.i.d. uniform U_i and arbitrary token-choice functions f_X, f_Y.

Proposition 7.3 (PTCRN $\le$ PTOC, prefix by prefix)**.** *Interpret $\mathbb{P}[X_j = a_j \mid X_{<j} = a_{<j}]$ as the model's next-token kernel, which is defined at every history, and similarly for Y. Under any PTCRN scheme, for every i and every a,*

$$\mathbb{P}[X_{<i} = Y_{<i} = a_{<i}] \le \prod_{j<i} \min\{\mathbb{P}[X_j = a_j \mid X_{<j} = a_{<j}],\ \mathbb{P}[Y_j = a_j \mid Y_{<j} = a_{<j}]\},$$

and PTOC attains equality.

Proof. Under PTCRN, X_j is conditionally independent of $Y_{<j}$ given $X_{<j}$. Indeed, $X_{<j}$ and $Y_{<j}$ are functions of $U_{<j}$, and $X_j = f_X(U_j, X_{<j})$ with U_j independent of $U_{<j}$. So given $U_{<j}$, the law of X_j depends only on $X_{<j}$, through the kernel $k_X(\cdot \mid X_{<j})$. Symmetrically for Y_j. Let $q_j = \mathbb{P}[X_{<j} = Y_{<j} = a_{<j}]$, and induct on $j \ge 1$, with $q_1 = 1$. If $q_j = 0$ then $q_{j+1} = 0$. Otherwise, conditioning on the event $\{X_{<j} = Y_{<j} = a_{<j}\}$, which has positive probability,

$$q_{j+1} = q_j \mathbb{P}[X_j = Y_j = a_j \mid X_{<j} = Y_{<j} = a_{<j}] \le q_j \min\{k_X(a_j \mid a_{<j}), k_Y(a_j \mid a_{<j})\},$$

because by conditional independence the conditional laws of X_j and Y_j given that event are $k_X(\cdot \mid a_{<j})$ and $k_Y(\cdot \mid a_{<j})$. PTOC has the same conditional independence and couples each step maximally, so equality holds at every step. □

Since the probability that the longest common prefix has length at least i is a sum of such prefix agreement probabilities, PTOC also dominates PTCRN for the length of the longest common prefix. Both this domination and that of PTOC by APOC are weak inequalities that can be strict for some distributions. For a particular pair of models, seed reuse can match PTOC: shared inverse-CDF sampling of binary tokens is already maximal, for example. But seed reuse in its usual form cannot match PTOC universally. There, each model's token-choice function depends only on that model's own next-token distribution, as with inverse-CDF sampling, so a single scheme couples any number of models at once. Three distributions need not admit a simultaneously pairwise maximal coupling [1]: uniform distributions on $\{a, b\}$, $\{a, c\}$ and $\{b, c\}$ are pairwise at distance $1/2$, and pairwise maximality would need three disjoint events of probability $1/2$. So for some pair the scheme is strictly worse than PTOC. APOC can be strictly better than PTOC:

Example 7.4 (PTOC is not maximal for whole responses)**.** Let $X \sim \mathrm{Unif}\{aa, ab, bb\}$ and $Y \sim \mathrm{Unif}\{aa, ba, bb\}$. A maximal coupling has $\mathbb{P}[X = Y] = 2/3$, for instance uniform on $\{(aa, aa), (ab, ba), (bb, bb)\}$, but PTOC achieves only $1/3$. Even with independent tokens, $\mathbb{P}[X_i = H] = 3/4$ and $\mathbb{P}[Y_i = H] = 1/4$ for $i = 1, 2$ give $1/4$ under PTOC but $1/2$ under a maximal coupling of the pair. Here the laminar coupling of Theorem 6.3 achieves $1/2$ for the pair while keeping each position maximally coupled.

8 Related work

Maximal coupling. Maximal couplings are credited to Doeblin [9]. For background see [10, 7]. The rejection-sampling algorithm is due to Johnson [6]. It is implicit in speculative decoding [8, 2], and the connection to optimal transport and coupling was made explicit by Sun et al. [14]. We do not know of an earlier description of MRC, that is, of using the input coupling's conditional sampler as the proposal in this rejection scheme.

Simultaneously maximal couplings of nested families. Griffeath [5] and Goldstein [4] constructed couplings of processes that are maximal for every tail σ-field at once. Völlering [16] and Ernst, Kendall, Roberts and Rosenthal [3] constructed maximal agreement couplings, which keep two processes equal for as long as possible. These are maximal for every prefix at once, as in Theorem 7.2. In speculative decoding, Sun et al. [15] showed that token-level verification is not optimal and proposed block verification. Their greedy variant corresponds to APOC. Makur and Singh [12] study simultaneously maximal couplings in connection with Doeblin coefficients, generalizing a result of Goldstein, and Polyanskiy and Wu [13] discuss when a joint law and its marginals can be maximally coupled at once. Angel and Spinka [1] study pairwise-optimal couplings of three or more variables.

Independence constraints. The question of which families can be simultaneously maximally coupled under independence or Bayesian-network constraints does not arise in Markov-chain coupling, where no such independence is present. That may be why we did not find it studied. Theorems 6.3 and 6.5 and the SMC-CPT algorithm are, as far as we know, new.

9 Open problems

Open problem 9.1 (infinite spaces)**.** Extend MRC to countable and continuous spaces, with densities in place of probabilities. The proofs here use finiteness in Lemma 4.2 through the uniform bound on exit probabilities, and in Proposition 3.11. The notes contain a sampling algorithm for a single chain step with infinite vocabularies, but no general version.

Open problem 9.2 (tails)**.** Theorem 3.2(d) bounds expected costs. In the finite case, the number L of loop iterations in one run has a discrete phase-type distribution. With $a = \alpha|_{R_X}^{\top}$ and $F = (I - Q_R)^{-1}$, we have $\mathbb{P}[L > k] = aQ_R^k\mathbf{1}$ and $\mathbb{E}[L^2] = a(2F^2 - F)\mathbf{1}$, and Lemma 4.2(ii) gives a geometric tail. These quantities depend on the whole flow, though. Give useful bounds on the variance or tail of L in terms of accessible parameters, such as $d_{\mathrm{TV}}(p_X, p_Y)$ and bottlenecks of the flow P, and for the nested calls of SMC-CPT.

Open problem 9.3 (characterizing MRC)**.** Example 3.10 shows that the properties in Theorem 3.2 do not determine $\hat{P}$. Is there a natural extra condition, for example a conditional-independence or Markov property of the shortcut decomposition, that singles out $\mathrm{MRC}(P)$, much as conditional-independence conditions single out APOC?

Open problem 9.4 (non-ancestral families and likelihood-free models)**.** Give efficient algorithms for the families of Theorem 6.5 that are not ancestral, using only the network's conditionals. Separately, MRC needs likelihoods. Can anything similar be done for samplers without likelihoods, such as diffusion models evaluated only through their per-step decisions?

Open problem 9.5 (applications)**.** Measure the gain of SMC-CPT over per-token coupling in practice, for example acceptance rates in speculative decoding of masked-diffusion models, or variance reduction when comparing two discrete-diffusion models.

10 Provenance and status

The research is Malcolm Sharpe's, carried out from February to October 2024 and recorded in blog posts that are archived alongside this document. The algorithms, the main theorems and most proof ideas come from those notes. This write-up was produced in October 2026 by an AI assistant (Claude Opus 5.5) at the author's request. It standardizes notation, completes the proofs and adds a few results. The proofs were reviewed by a second model (GPT-6 Astra), and the main claims were checked numerically: floating-point evaluation of the finite-state formulas on random instances, supplemented by Monte Carlo simulation of the literal algorithm. Relative to the notes:

- The Markov-chain facts the notes left as beliefs (recurrence, Cesàro limits, return times) are replaced by Lemmas 4.1 and 4.2. They need only flow conservation and a spectral-radius bound, and they cover the reducible case.
- The pseudoinverse form of the closed formula, which the notes marked as low confidence, is proved (Proposition 3.4).
- The conjecture that SMC on autoregressive models reproduces APOC is proved (Theorem 7.2).

- The characterization theorems' "only if" directions are proved in general from the notes' counterexamples (Theorems 6.2, 6.3 and 6.5).
- Corrections: the notes' pseudocode for the tensor product called a Y-sampler where an X-sampler was meant. One note stated the total-variation bound with the sign of the gain reversed. One example meant to show a maximal coupling moving by total variation 1 listed the same coupling twice (Example 3.9 gives a correct version).

Errors may remain. Please treat the novelty assessments in particular as provisional.

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A Glossary: terminology of the original notes

The archived notes changed terminology over time. This table maps their terms to the ones used here.

Notes	This document
optimal coupling; maximum coupling	maximal coupling
CMC, classic maximal coupling	Algorithm 1
MRC, maximal recoupling; MRC-Cond	Algorithm 2
APOC, all-prefix-optimal coupling; APMC; SMC-AR	Π_n, Section 7
PTOC, PTC (per-token (optimal) coupling)	PTOC
PTCRN, CRN, seed reuse	PTCRN
NCFOC (non-crossing-family-optimal coupling)	simultaneously maximal coupling of a laminar family under independence (Theorem 6.3)
non-crossing family	laminar family
SMCBN, SMC-BN; SMC-I	Theorem 6.5; Theorem 6.3
SMC-CPT; Tree (Leaf, Split, Step); TensorProduct; ChainRule	SMC-CPT; CPT (leaf, product, chain); $\otimes$; $\ltimes$
loss of a coupling	$\sum_a \min\{p_X(a), p_Y(a)\} - \mathbb{P}[X = Y]$
cost decrease; equality-probability improvement	γ_f
subcoupling; equivalence classes $[\cdot]$	coarsening P^f; fibres of f
irreducible P	every node with $p_{\max} > 0$ reaches a sink (U contains no arcs)
$g_{\text{old}}(t)$, $g_{\text{new}}(t)$	$g(t)$, $\hat{g}(t)$

B Implementation-level pseudocode

A joint object J over a set E provides `LX`, `LY` (marginal likelihoods), `sampX()` and `sampYgivenX(x)`. CPT nodes are `Leaf(mX, mY)`, `Product(t1, t2)` and `Chain(t, cmX, cmY)`, where `m` is a marginal oracle and `cm` a conditional oracle. This corrects the pseudocode in the notes of 2024-10-08 and 2024-10-10.

```
Build(Leaf mX mY)      = MRC(Independent(mX, mY))
Build(Product t1 t2)   = MRC(Tensor(Build t1, Build t2))
Build(Chain t cmX cmY) = MRC(ChainRule(Build t, cmX, cmY))

Independent(mX, mY):
  LX(x) = mX.L(x);  LY(y) = mY.L(y)
  sampX()          = mX.samp()
  sampYgivenX(_)   = mY.samp()

Tensor(j1, j2):                        # requires independence of the two scopes
  LX((x1,x2)) = j1.LX(x1) * j2.LX(x2)
  LY((y1,y2)) = j1.LY(y1) * j2.LY(y2)
  sampX()               = (j1.sampX(), j2.sampX())
  sampYgivenX((x1,x2))  = (j1.sampYgivenX(x1), j2.sampYgivenX(x2))

ChainRule(j, cmX, cmY):
```

```
  LX((x1,x2)) = let a = j.LX(x1) in (a == 0 ? 0 : a * cmX.L(x1, x2))
  LY((y1,y2)) = let b = j.LY(y1) in (b == 0 ? 0 : b * cmY.L(y1, y2))
  sampX()               = let x1 = j.sampX() in (x1, cmX.samp(x1))
  sampYgivenX((x1,_))   = let y1 = j.sampYgivenX(x1) in (y1, cmY.samp(y1))

MRC(j):
  LX = j.LX;  LY = j.LY;  sampX = j.sampX
  sampYgivenX(x):
    with prob min(1, j.LY(x) / j.LX(x)): return x
    v = x
    loop:
      w = j.sampYgivenX(v)
      with prob min(1, j.LX(w) / j.LY(w)): v = w; continue
      return w

Sample(tree): j = Build(tree); x = j.sampX(); return (x, j.sampYgivenX(x))
```

In practice, compute likelihoods in log space, since products of many conditionals underflow. Cache likelihoods of points already visited.